Fiscal Year:	FY 2019	**Task Last Updated:**	FY 09/02/2018
PI Name:	Govind, Shubha Ph.D.		
Project Title:	Does Spaceflight Alter the Virulence of a Natural Parasite of Drosophila?		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	New York	**State:**	NY
Zip Code:	10031-9101	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 10/31/2019 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 10/31/2018 per F. Hernandez/ARC (Ed., 9/21/17) NOTE: End date changed to 10/31/2017 per NSSC information (Ed., 8/3/16)		
Key Personnel Changes/Previous PI:	September 2018: (1) A new graduate (PhD) student Mr. Waleed Khalid joined the project in March 2018. (2) Mr. Jackson Chen's appointment is ending at the end of August.		
COI Name (Institution):	Bhattacharya, Sharmila Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	NNX15AB42G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The seminal discovery of Toll and Toll-NF-KB signaling as a central mediator of innate immune responses in the model insect Drosophila catalyzed similar discoveries in other organisms including innate immunity in humans. These findings led to an appreciation of the fundamental roles of innate immunity in human immune deficiency on the one hand, and immune health of insects on the other. In Drosophila larvae, primed, but quiescent hematopoietic progenitors are called into action from an organ with functional similarities to mammalian bone marrow. Blood cells signal the fat body (equivalent to the mammalian liver) to activate the expression of antimicrobial peptide genes (humoral immunity). Antimicrobial peptides efficiently combat deadly bacteria and fungi. The Leptopilina wasps are natural metazoan enemies of Drosophila and their attack kills fruit flies. Wasp attack (e.g., L. boulardi) activates both the humoral and cellular immune arms of larval hosts, each being dependent on Toll-NFKB signaling. Previous NASA-funded studies of Drosophila have shown that stresses in the space environment compromise hematopoietic development and modulate larval and adult innate immunity. We hypothesize that stimuli from the space environment will also alter the virulence of the parasitic wasps, and therefore their success on the fly hosts cultured in space. In Drosophila-wasp co-culture experiments using the NASA Fruit Fly Lab hardware over 30-days, we propose to (1) examine whether the fly or the wasp partner wins the arms race in space; (2) examine the effects of spaceflight on the anti-parasite response in fruit fly hosts; and (3) examine if virulence of "space-cultured" wasps is different from counterparts grown on Earth. While both L. boulardi and L. heterotoma parasites always succeed on the wild type, only the latter wasp species produces unique cell-killing activities residing in its virus-like particles. It also suppresses humoral immunity completely. We will examine if changes in the expression of VLP (virus-like particles) genes underlies changes in parasite virulence affected by spaceflight. Host defense and virulence will be inferred from a suite of other well-established assays of fly/host survival and measures of immune response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Parasitic wasps of Drosophila are a powerful and increasingly well-characterized model for understanding longstanding questions about natural host/parasite interactions, allowing molecular genetic examination of both innate immune mechanisms in fly hosts and virulence mechanisms of their wasp parasites. Distant relatives of these parasitic wasps are used for biocontrol of agricultural pests worldwide. The "Earth benefits" impact of our research over the last yeast has been as follows: (a) Successful culture of parastic wasps on their Drosophila melanogaster hosts was achieved in the mission. Leptopilina heterotoma cultures were more successful than L. boulardi cultures, likely due to their high virulence. This experiment revealed that these insects can undergo normal development even in the adverse conditions of the international space station. (b) Genomic sequences of male and female L. heterotoma wasps were obtained and assembled to study the gene structures and expression of several virulence genes. These foundational results will help in prioritizing the genes whose expression will be studied in detail in space/ground samples. They will also help annoation and assembly of the genomic sequences. Overall, our studies open new areas of research in the area of infectious disease and human health. They strengthen the Drosophila host-parasite model system that has served as a stand-in for human and vertebrate studies on Earth and in space.
Task Progress:	During the last reporting period the following progress was made: 1. Extensive testing to optimize conditions of culture for flight and ground controls was performed. Fly stocks and food was shipped out from Principal Investigator (PI)'s lab to NASA team at Ames for preparation of EVT (experiment verification test) in September 2017. EVT was conducted in November-December of 2017. 2. Preparation of flight continued through launch date of April 2, 2018. During this period, cultures of the animals were expanded in number taking care that there was no contamination. 3. In late Feb of 2018, these cultures were shipped to Kennedy Space Center (KSC). 4. Pre-launch preparations occurred in March 2018. 5. Post-launch experiments were conducted in May and June of 2018. This included direct observations of animals from the flight as well as storage of samples for subsequent experiments. 6. Two manuscripts were submitted for publication over the summer of 2018. 7. Three presentations will be made in 2018-2019. Documents for these presentations were sent to conferences. 8. A number of outreach efforts were made after the launch of the experiments. 9. One PhD student graduated in May 2018 and another joined the lab in March 2018. 10. One undergraduate student graduated at the end of summer of 2018. 11. One high school is working on an internship project since summer of 2018.
Bibliography Type:	Description: (Last Updated: 01/09/2024)
Abstracts for Journals and Proceedings	Heavner ME, Ramroop J, Razzak Z, Wey B, Govind S. "Parasitic wasp-derived multi-strategy extracellular vesicles specifically interact with cells of the Drosophila host's cardiovascular/hematopoietic systems." Presented at the American Society of Cell Biology Annual Meeting, Philadelphia, PA, December 2-6, 2017. Immune System Session. Molecular Biology of the Cell. 2017 Dec;28(26):Poster No. P1858. https://doi.org/10.1091/mbc.e17-10-0618 ; PubMed Central PMCID: PMC5739290 (Go to Supplementary Material link) , Dec-2017
Abstracts for Journals and Proceedings	Wey B, Ramroop J, Heavner ME, Razzak Z, Scarpati M, Singh SM, Govind S. "A T3SS SipD-like protein aids parasite success in Drosophila." RCMI Translational Science 2017 Conference, Washington DC, October 28-November 1, 2017. RCMI Translational Science 2017 Conference Abstract # 01.07.005. , Oct-2017

Abstracts for Journals and Proceedings	Jerliu A, Heavner ME, Govind S, Singh S. "Computational modeling and characterization of Diedel proteins that regulate highly-conserved immune signaling." American Association for Cancer Research Annual Meeting 2017, Washington, DC, April 1-5, 2017. Cancer Research. 2017 July;77(13 Suppl):4537. https://doi.org/10.1158/1538-7445.AM2017-4537 , Jul-2017
Abstracts for Journals and Proceedings	Heavner ME, Ramroop J, Wey B, Crissman J, Miller E, Singh S, Govind S. "A new family of GTPases from virulence-linked extracellular microvesicles of a Drosophila generalist parasite." 59th Annual Drosophila Research Conference, Philadelphia, PA, April 11-15, 2018. Annual Drosophila Research Conference 50 Genetics Society of America, 2018. Pgm Nr 131. , Apr-2018